

Transformer Encoder for Text Classification:

Implementation, Ablation Analysis, and Attention Validation on SST-2

Introduction

Over the last decade, there has been an epoch-making shift within natural language processing, with the main driver being the release of the Transformer architecture proposed by Vaswani et al. in the famous Attention Is All You Need paper in 2017. Before the invention of the Transformer, the most common approach to processing text sequentially utilized recurrent neural network architectures such as Long Short-Term Memory networks and its variations. Recurrent neural networks operate on input tokens sequentially and thereby introduce two key disadvantages: firstly, sequential computation does not allow for parallel training across positions; secondly, it makes it difficult to model dependencies that occur in distant parts of a text as information has to traverse multiple sequential computational steps to reach distant positions.

The basic idea behind self-attention, which is at the heart of Transformers, is to ask three specific questions for every position within a sequence of elements. These include what type of information this particular position seeks, which is described using a query vector; what information is available at each particular position, which is described using a key vector; and what kind of content this particular position has to offer, which is represented using a value vector. To perform this operation, the model calculates the compatibility score between each position's query vector and the key vector of every position. These compatibility scores are then converted into probabilities using the softmax function. These probabilities are used to calculate a weighted average of the value vectors from every position within the sequence. This enables every position within a sequence to attend to every other position simultaneously without any concern regarding the distance involved. Mathematically, $\text{Attention}(Q, K, V) = \text{softmax}(QK^T/\sqrt{d_k})V$, where

Q, K, and V represent the matrix containing queries, keys, and values. These vectors are obtained by applying learned linear transformations to the inputs, while d_k represents the dimensions of the keys.

This project attempts a complete from-the-ground-up implementation of the Transformer encoder in binary sentiment classification on the Stanford Sentiment Treebank SST-2 corpus using only the tensor operations of PyTorch along with its autograd capabilities, foregoing any use of pre-existing transformer modules. The rationale behind this project is to achieve an in-depth understanding of all components of the model. This study compares the Transformer against two baselines: the classic TF-IDF + logistic regression combination that represents the pre-neural era, and the bidirectional LSTM that represents the pre-attention sequential modeling era. There are three research questions driving this study. The first question is whether the learned attention-based representation outperforms the TF-IDF-based feature weight in terms of accuracy and importance of the words used. The second question is whether the attention heads specialize in certain linguistic roles. The third question is whether the attention weights have causal significance on the prediction or are merely correlated with the prediction.

This project builds upon the interpretability analysis conducted by Clark et al. (2019) in their paper titled What Does BERT Look At? An Analysis of BERT's Attention, which was presented in the workshop on BlackboxNLP of the 2019 ACL conference. In this analysis, Clark et al. examine BERT's 144 attention heads using three approaches: visualization of attention weights directly, analysis of surface patterns based on what tokens each head pays attention to, and probing for syntactic dependency relations using classification probes. The main result is that most of the heads acquire a consistent behavior, either looking at adjacent tokens or more specifically at syntactically related tokens, like direct objects and subject nominatives, even though BERT was not trained for

such tasks. Some heads reach an accuracy of over 70 percent when identifying particular dependency relationships between words, while a simple logistic regression probe using just the attention weights achieves state-of-the-art results when classifying dependency arcs, thus proving that the information about syntactic structures is represented linearly in the attention mechanism.

It also considers another issue that has not been addressed by Clark et al. – whether or not attention weights play a role as an explanatory factor when deciding what to focus on in a particular input. As was argued by Jain & Wallace (2019), there was little connection between attention and gradient-based importance, yet Wiegrefe and Pinter (2019) proved that there were certain situations where attention could act as a causative factor. To quantify this, this project uses a Spearman rank correlation method and calculates attention weights compared to the gradient saliency score, and carries out the attention faithfulness experiment where top-attention tokens are being replaced with random tokens and then measuring the difference in performance.

Methods

The main model architecture is a Transformer encoder used for binary classification of input sentences, trained from scratch using only PyTorch tensor operations, without any transformer libraries. The datasets employed include the Stanford Sentiment Treebank SST-2 benchmark proposed by Socher et al. (2013). This benchmark consists of sentences extracted from movie reviews and labeled as positive or negative, with a total of 67,349 training samples and 872 validation samples. The vocabulary has been created based on the training dataset only, containing the top 20,000 most frequently occurring word-level tokens. After filtering out rare tokens, the vocabulary consisted of 14,735 distinct tokens. In addition to that, three extra tokens were included in the vocabulary: a padding token, an unknown token, and a classification token.

Transformer encoder operates using six steps. In the first step, the input tokens are encoded using an embedding lookup table to dense vectors and are then scaled using $\sqrt{d_{\text{model}}}$, where d_{model} is the value of 256 for the embedding size. In the second step, sinusoidal positional encodings are applied to the scaled embeddings. The positional encodings at a certain position pos and index i of a dimension are calculated by $\sin(\text{pos} / 10000^{(2i/d_{\text{model}})})$ if the index is even and $\cos(\text{pos} / 10000^{(2i/d_{\text{model}})})$ otherwise. These sinusoidal encodings have a key geometrical property that makes them useful: they encode the information about position $\text{pos} + k$ using the linear transformation of position pos , which means that relative distance relations can be learned from them. In the third step, the encoded vector is fed through four identical stacked encoder blocks, with each one consisting of the self-attention mechanism and the feedforward neural network sublayers, separated by residual connections and normalization layers. There are four parallel attention heads, processing 64-dimensional subspaces of the overall 256-dimensional space. Inner dimension of feed-forward sub-layer is 1024 based on four times d_{model} as described by Vaswani et al. Layer normalization normalizes activations along the feature dimension for each position and performs scaling and shifting using learned parameters. Fourth, hidden state of classification token after passing through the last layer of encoder network is fed into a linear classifier having two output units. Total number of parameters of the entire model is 6,927,618 and Xavier uniform initialization is performed on all parameters.

The training algorithm utilizes the Adam optimization function with the Vaswani et al. learning rate warmup method. Since the warmup method explicitly controls the learning rate, the Adam optimizer is initialized with a base learning rate of 1.0, and the LambdaLR scheduler is applied with the expression d_{model} to the power of negative one-half multiplied by the minimum value of step to the power of negative one-half or step times 400 to the power of negative three-halves

with 400 warmup steps. The Adam betas and epsilon are 0.9, 0.98, and 10 to the power of negative 9, respectively, which match the values in Vaswani et al. precisely. Label smoothing with a value of 0.1 replaces the target labels of 0 and 1 with 0.05 and 0.95. Gradient norm is set to 1.0. Dual criterion early stopping occurs after 6 epochs if the improvement in validation accuracy becomes less than 0.001 per epoch, or when the difference between training and validation accuracies surpasses 12% for 4 continuous epochs; the best saved checkpoint is then loaded. Exponential moving average of weights is used starting epoch 8 with 0.99 decay.

Adam optimizer with a cosine learning rate decay from 10^{-3} to a minimum of 10^{-5} over the number of maximum training epochs, which is 20, is used in the Bidirectional LSTM baseline. Bidirectional LSTM employs a label smoothing of 0.05, weight decay of $5 \cdot 10^{-4}$, and dropout of 0.5. The Bidirectional LSTM has stricter criteria for early stopping that include patience threshold of three epochs and maximum training/validation gap of 10%. The baseline method of using TF-IDF with logistic regression involves representation of each sentence using a sparse vector with features based on the top 20,000 unigrams and bigrams with log-term frequency and inverse-document frequency scaling fitted using logistic regression where the regularization coefficient is 1.0.

There are four versions of the ablation where the only difference between the version and the complete model is the exclusion of a specific property. This means that the four ablation methods remove either the positional encoding, replace the four-headed attention mechanism with the single-headed version acting on the d_{model} -sized input, omit all residual connections, or exclude all layer normalization layers. The five versions have exactly the same hyperparameter settings, random seeds of 42 for the initial weights, and an early stopping strategy.

Three experiments in interpretability are conducted on the trained model. The probes used in probing classifiers are logistic regression models which are trained using the frozen encoder's hidden states at each layer in order to make predictions of the following three linguistic features: Sentiment polarity at classification token positions, Identity of negation words at each token positions, and Identity of intensifier words at each token positions. In each case, 80% of the validation dataset is used to train the model while 20% is used to evaluate it. The gradients saliency maps are computed by calculating the derivative of the logit predicted class with respect to the token embeddings and taking the L1 norm of the element-wise product of the embedding and its gradients. This map is normalized to have values ranging from zero to one. Attention validation experiment measures Spearman Rank correlation between the average CLS attention weights of all the layers and heads with the gradients saliency scores for 200 validation examples per layer of the encoder. Further, it checks whether removing top-K attention tokens results in bigger drop in accuracy than removing K randomly selected tokens in the input, averaged over 10 random trials.

All experiments have been conducted using Python, PyTorch v2.0 or newer, the HuggingFace datasets package to load SST-2, and scikit-learn for TF-IDF, logistic regression, and probing classifiers. The visualizations have been created using matplotlib and seaborn. The training was performed using an NVIDIA GeForce RTX 4090 Laptop GPU.

Results

The main classification results for all three models over the SST-2 validation dataset are listed in Table 1. Regarding the Encoder of the Transformer Model, we have an accuracy rate of 80.85%, Macro F1 score of 0.8085, MCC score of 0.6181, and AUC-ROC of 0.8869. Regarding the Bidirectional LSTM, the statistics obtained are an accuracy rate of 82.57%, Macro F1 score of

0.8254, MCC score of 0.6516, and AUC-ROC of 0.9012. Finally, in terms of the Baseline Model that utilized the TF-IDF and Logistic Regression combination, the scores were 8

Model	Accuracy	F1 Macro	F1 Wtd	MCC	AUC-ROC	Best Ep.
TF-IDF + LR	80.96%	0.8088	0.8091	0.6211	0.9065	N/A
BiLSTM	82.57%	0.8254	0.8255	0.6516	0.9012	6
Transformer (from scratch)	80.85%	0.8085	0.8084	0.6181	0.8869	7

Table 1. Evaluation Metrics for all three classifiers based on SST-2 Dataset (872 samples). F1 Macro: Simple average of the F1 score of each class. MCC: Matthews Correlation Coefficient whose value lies between -1 and 1 with -1 being the worst score and 1 the best one. Best Epoch: the epoch with the highest accuracy of the model.

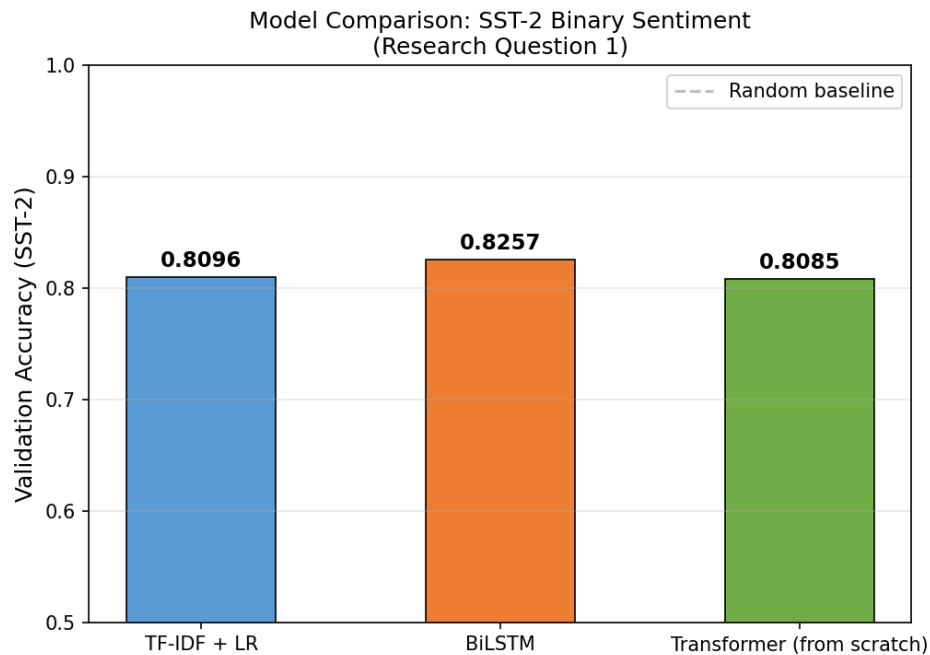


Figure 1. Validation accuracy comparison across the three models on the SST-2 validation set.

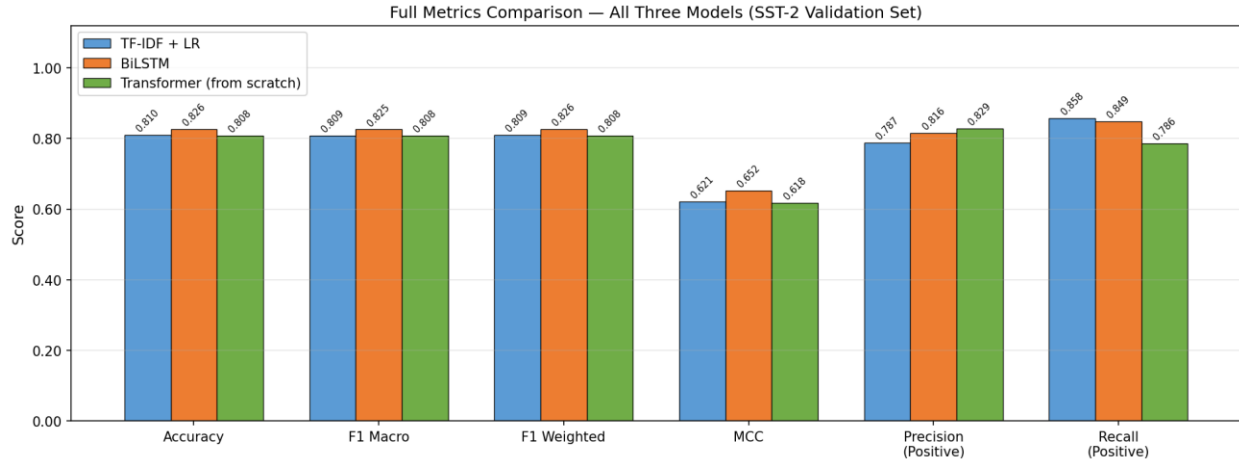


Figure 2. Full six-metric comparison across all three models including Accuracy, F1 Macro, F1 Weighted, MCC, Precision-Positive, and Recall-Positive.

For our task, the best performing algorithm is the BiLSTM with an accuracy of 82.57%, followed by the TF-IDF baseline with accuracy of 80.96% and the Transformer at 80.85%. One interesting point here is the systematic mismatch between the precision and recall of the TF-IDF negative class (0.8376 and 0.7593, respectively) due to its inability to properly evaluate negated sentences and the correct classification of only 325 out of 428 negative samples, whereas it obtains a high recall of the positive class (0.8581) as a result of its tendency to rely only on the globally high-weight positive sentiment words. At the same time, the Transformer produces fairly similar metrics for positive classes with a slightly higher negative precision (0.7894 vs 0.8290) but much better recall (0.8318 vs 0.7860), demonstrating how the attention mechanism is capable of adjusting word weights according to the surrounding context. Finally, the BiLSTM yields the best per-class performance with negative precision and recall of 0.8366 and 0.8014 and positive precision and recall of 0.8160 and 0.8491. As for the AUC-ROC metric, the best performer is the TF-IDF with 0.9065 as a result of its highly calibrated probability estimation through logistic regression.

Training dynamics of the Transformer indicate well-balanced training runs. The peak learning rate on the warmup schedule was approximately 1.36 multiplied by 10 to the power of negative 3, at which point the learning rate dropped in smooth fashion throughout the subsequent epochs. The best checkpoint of the model was found in epoch 7 with validation accuracy equal to 80.85%. The process of early stopping started in epoch 13, due to six successive epochs without improvements in validation accuracy. The maximum training/validation accuracy difference amounted to 9.3% at epochs 8 and 9 and was lower than 12% at all epochs. As for the BiLSTM model, the learning rate was smoothly decreased through a cosine annealing scheduler, starting with the rate of 10 to the power of negative 3 until reaching the minimum of 10 to the power of negative 5. The checkpoint with the best accuracy was obtained at epoch 6 with 82.57% validation accuracy and stopped at epoch 9 due to no improvements in validation accuracy for three successive epochs. The gap reached the value of 10.2% at epoch 9.

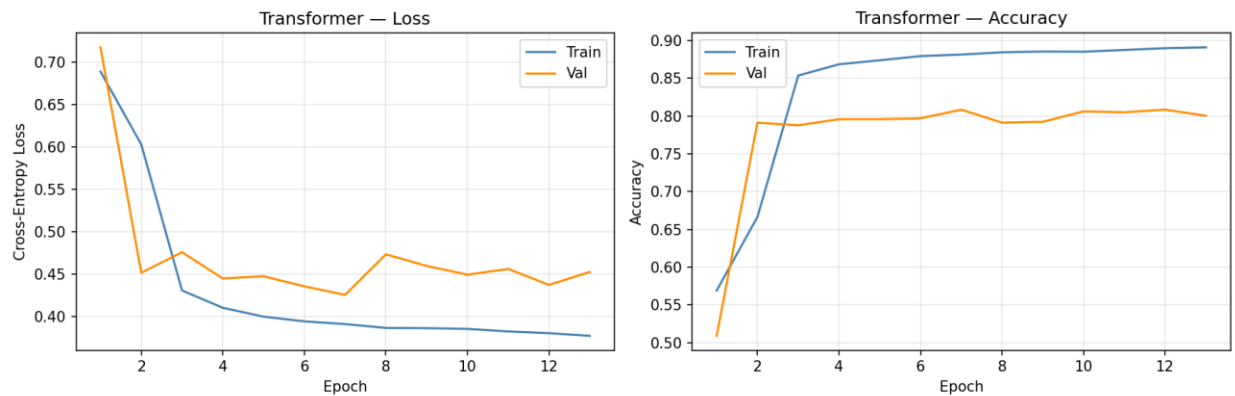


Figure 3. Transformer encoder training and validation accuracy (left) and loss (right) across 13 training epochs before early stopping. The Vaswani warmup schedule is visible in the initial learning rate behavior.

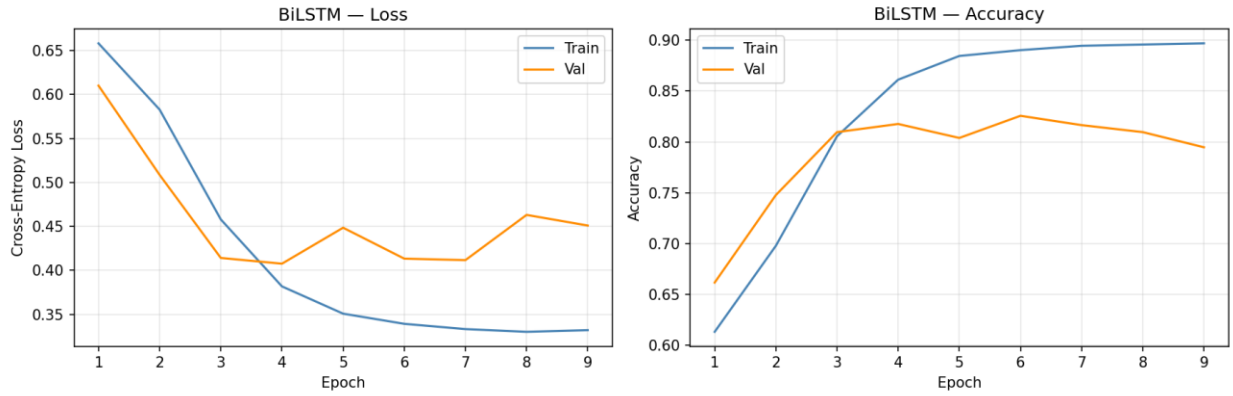


Figure 4. BiLSTM training and validation accuracy (left) and loss (right) across 9 training epochs before early stopping. Cosine annealing LR decay produces a smooth convergence pattern.

All three confusion matrices can be seen in Figures 5, 6, and 7. In the case of the TF-IDF model, out of 428 negative samples and 444 positive samples, respectively, 325 negative samples and 381 positive samples were correctly classified, resulting in 75.9% True Negative Rate and 85.8% True Positive Rate. For BiLSTM, the model correctly classified 343 negative samples and 377 positive samples. Finally, with the Transformer model, there was correct classification of 356 negative samples and 349 positive samples.

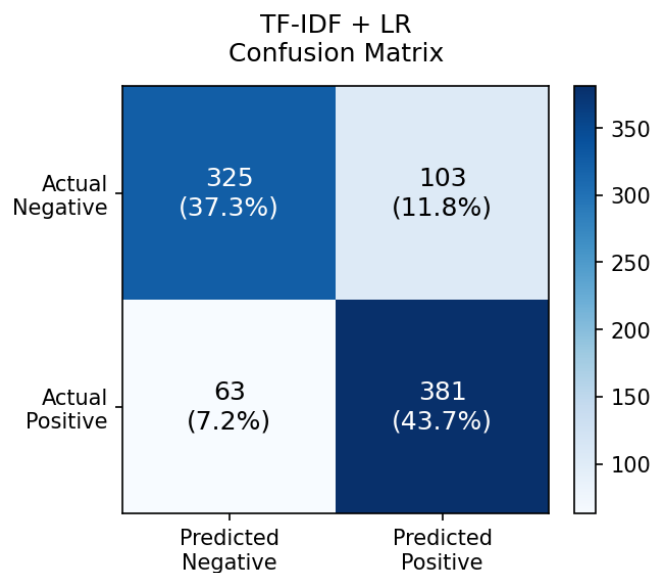


Figure 5. Confusion matrix for the TF-IDF + Logistic Regression baseline showing $TN=325$, $FP=103$,
 $FN=63$, $TP=381$.

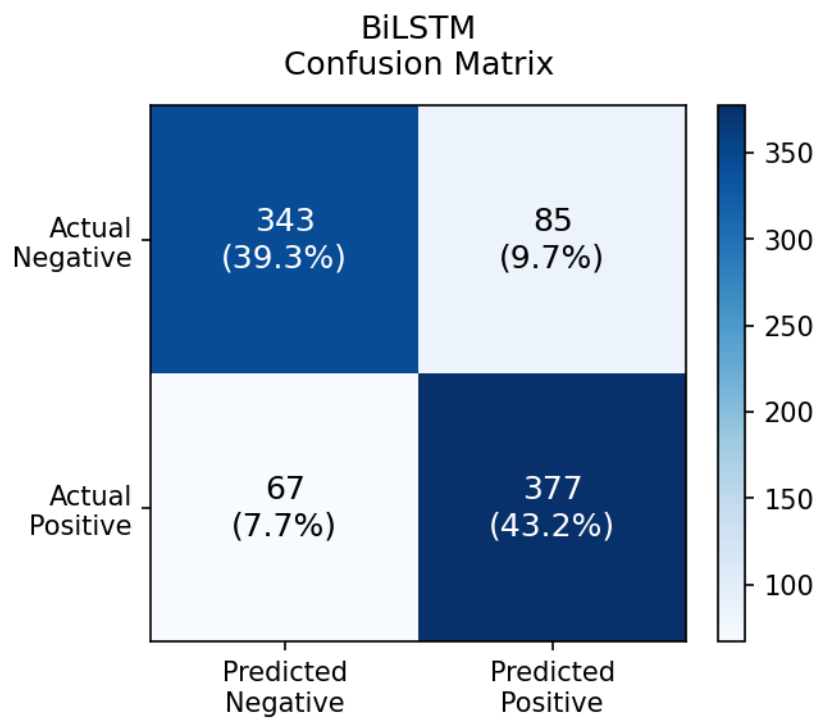


Figure 6. Confusion matrix for the Bidirectional LSTM baseline showing $TN=343$, $FP=85$, $FN=67$,
 $TP=377$.

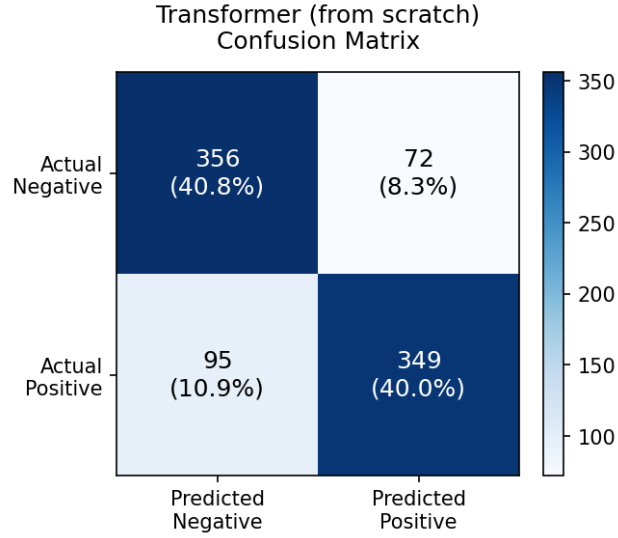


Figure 7. Confusion matrix for the Transformer encoder showing $TN=356$, $FP=72$, $FN=95$, $TP=349$.

The ablation study findings are presented in Table 2. The most surprising result is the failure of the network without residual connections, which attains a mere accuracy of 50.92% and an MCC of 0.000. This implies that the model classifies all samples from the validation dataset as positive, hence having 0 true negatives and 428 false positives. This experiment shows that residual connections are necessary rather than just optional since without identity skip connections in each sublayer for calculating gradients, gradient vanishing occurs in the early encoder layers, and therefore, no learning occurs, except for a majority class classifier after 7 training epochs. Without positional encoding, the network achieves 81.42% accuracy, which is a marginal increase of 0.46 percentage points compared to the full network. This experiment reveals that explicit position information is unnecessary for SST-2 sentiment analysis. In single head attention, the accuracy drops by 0.57 percentage points to 80.39%, while it takes 20 epochs to achieve the best checkpoint compared to 11 for the full model. Without the use of layer normalization, the accuracy reaches 81.31%, an improvement by 0.35%, but the epoch one training loss at 2.676 is more than four times that of all the other variants tested.

Variant	Accuracy	F1 Macro	MCC	AUC-ROC	Δ Acc	Stopped Ep.
Full Model (baseline)	80.96%	0.8096	0.621	0.876	—	11
No Positional Encoding	81.42%	0.8139	0.629	0.896	+0.46%	15
Single-Head Attention	80.39%	0.8038	0.611	0.877	−0.57%	20
No Residual Connections	50.92%	0.3374	0.000	0.500	−30.04%	7
No Layer Normalization	81.31%	0.8127	0.627	0.890	+0.35%	9

Table 2. Results of ablation study. Δ Acc = difference in accuracy compared to baseline (full model). All models have: d_model=256; four encoder layers; all hyperparameters; and random seed 42. Without Residual Connections leads to total failure of the model (MCC=0.000, true negatives = 0).

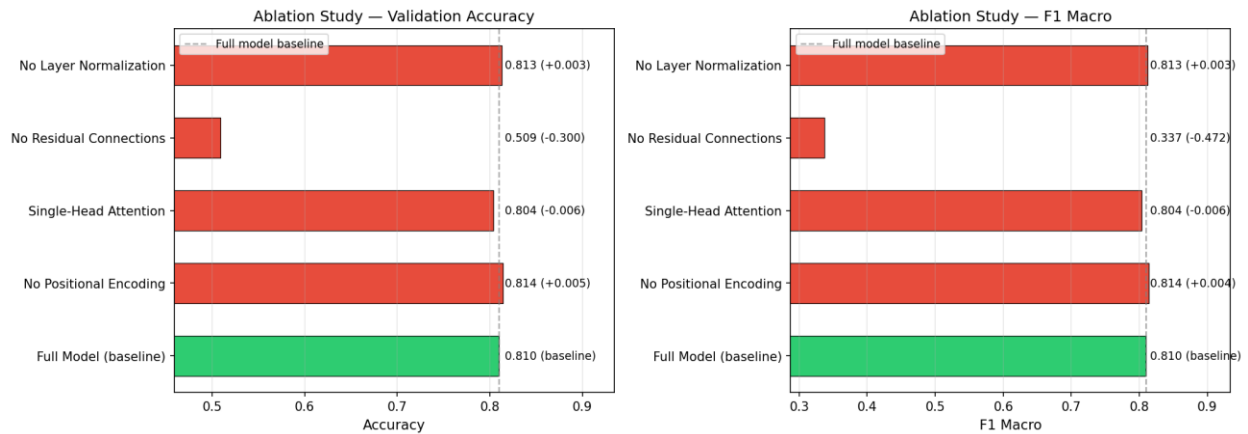


Figure 8. Ablation study accuracy and F1 macro comparison across all five variants, with delta annotations showing change from the full model baseline.

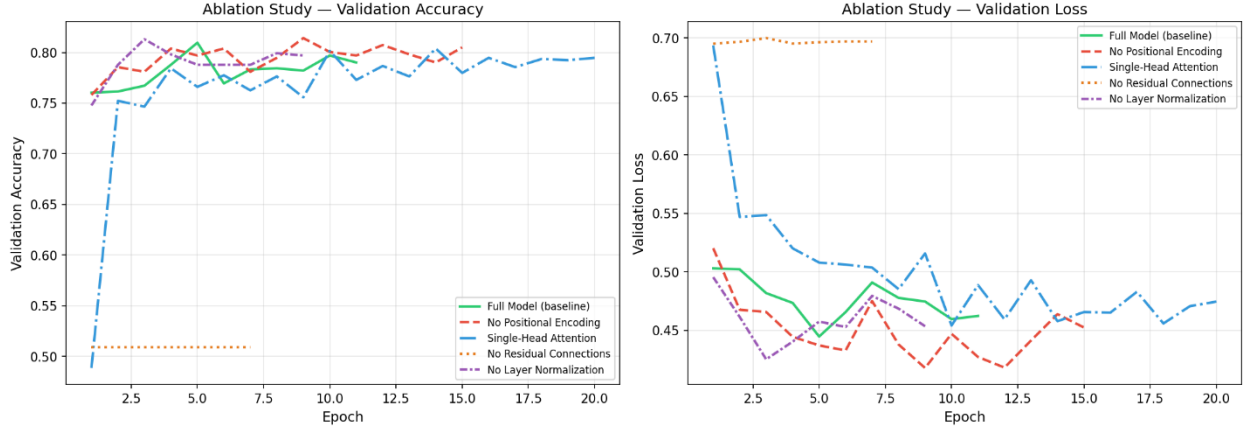


Figure 9. Curves for the validation accuracy with respect to the number of epochs of training for all five variations. In case of the variation with no residual connections, there is a horizontal curve around 51%.

The results of the probing classifier experiments are shown in Table 3. The sentiment polarity is decodable from all four encoder layers at more than 79%, including 80.0% for Layer 1, 80.6% for Layer 2, 80.6% for Layer 3, and 79.4% for Layer 4. The high and consistent decoding accuracy rates from all the four layers imply that sentiment signals have been preserved even at the very first layer, although there is some additional sentiment-related computation going on in the first two attention modules. Decoding results for the negation word identity exhibit a pronounced monotonic decrease: 62.3% for Layer 1, 58.0% for Layer 2, 56.8% for Layer 3, and 47.1% for Layer 4. The fact that negation identity decreases from 62.3% at Layer 1 to subchance level at Layer 4 implies that the token identity is increasingly distributed across the hidden states of the token that negates another token through the attention operation. Word identities of intensifiers are relatively consistent throughout the layers, ranging from 59.1% to 60.1%. This implies that the word identities of intensifiers are not highly localized nor redistributed through the attention mechanism within the corpus.

Probe Target	Layer 1	Layer 2	Layer 3	Layer 4
Sentiment polarity (CLS token)	80.0%	80.6%	80.6%	79.4%
Negation word identity	62.3%	58.0%	56.8%	47.1%
Intensifier word identity	59.1%	59.1%	60.1%	58.9%

Table 3. Testing classifier accuracy in every encoder layer. The accuracy of the balanced majority-class is about 50% for all token probes due to balancing of classes. The probe of sentiments is based on CLS token position while the probes of negations and intensifiers are based on tokens’ positions separately. Every probe was trained using 80% of validation hidden states and evaluated with 20%.

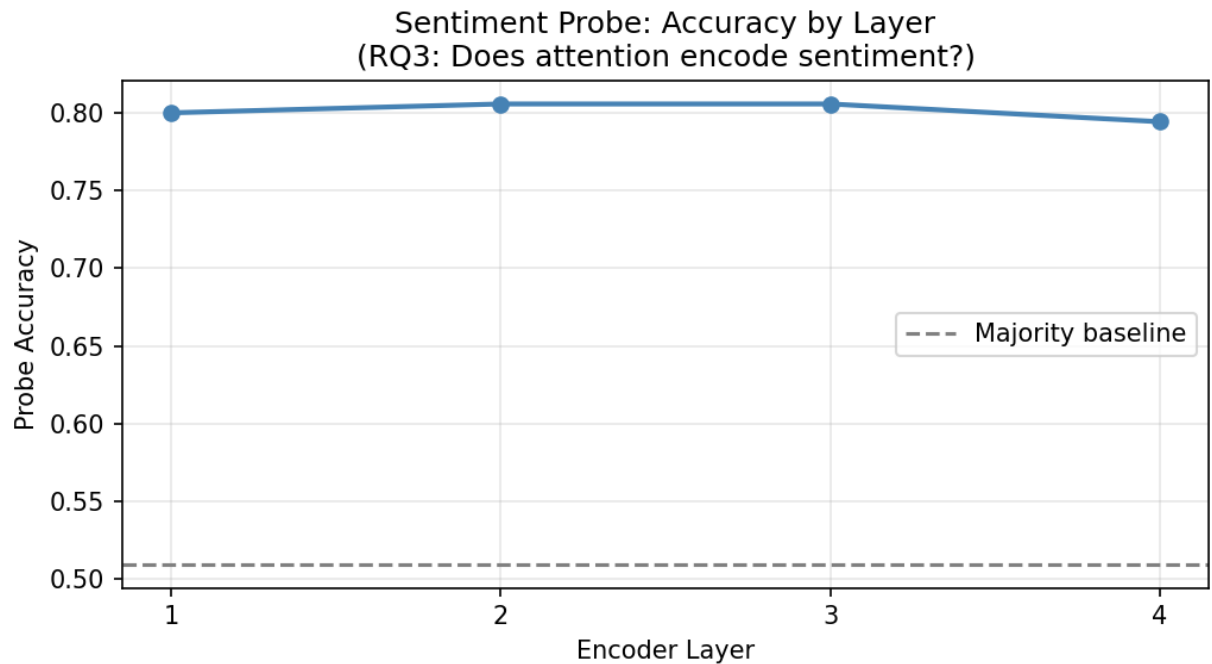


Figure 10. Sentiment probing accuracy by encoder layer with the majority-class baseline shown as a dashed gray line.

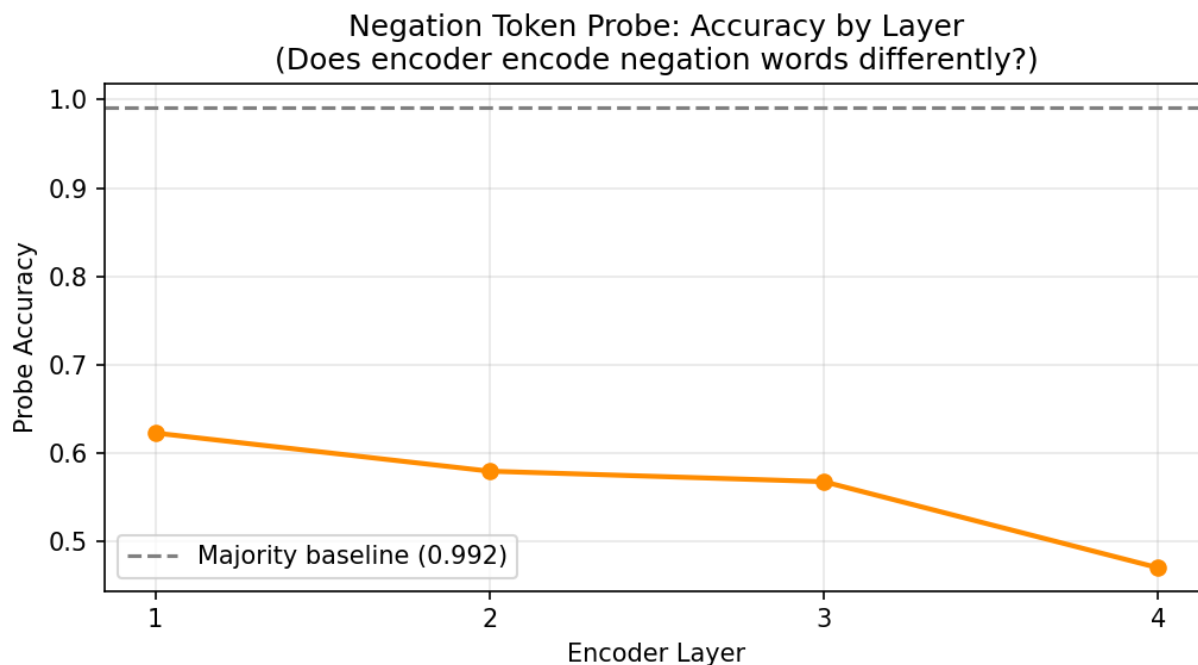


Figure 11. Negation word probing accuracy by encoder layer. The monotonic decline from 62.3% to 47.1% reflects attention-based redistribution of negation information.

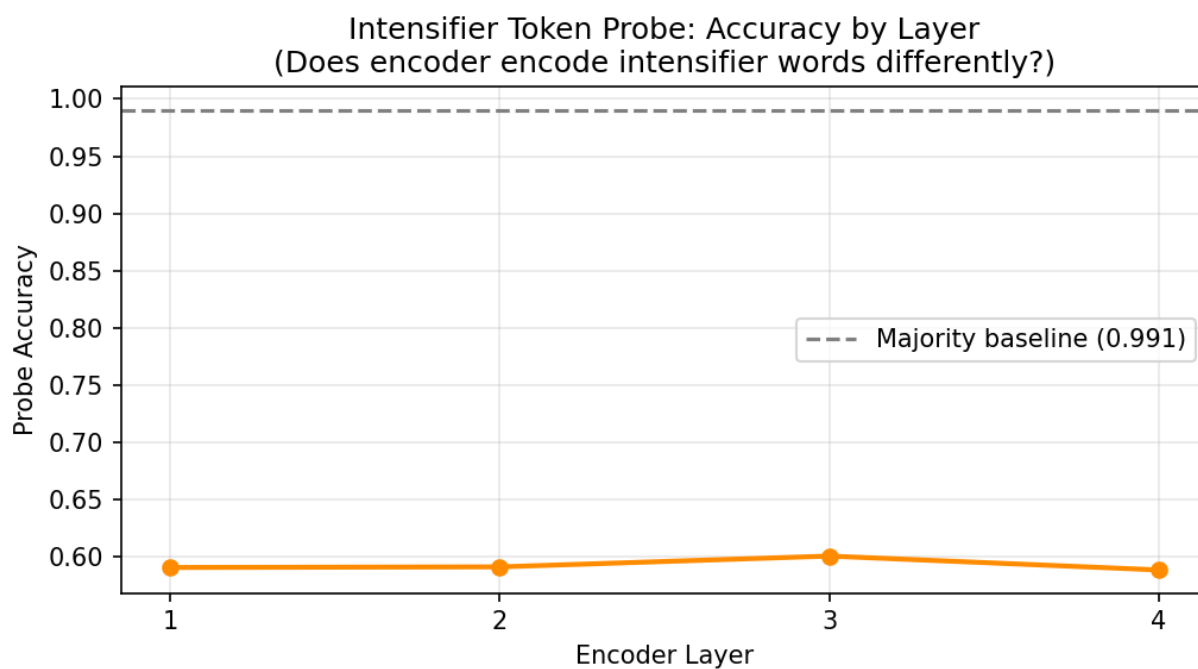


Figure 12. Intensifier word probing accuracy by encoder layer showing relative stability across all four layers.

The attention visualization results are depicted in figures 13 to 16 below. It can be seen from the head summary across 20 batches in validation that the Layer 1 heads have low Shannon entropy and high adjacent-token attention weight, acting as contextual aggregation mechanism like the sliding window method. Layers 3 and 4 are increasingly more diffuse and focused on each sentence's specific attention and increasing their focus on the classification token position. The attention heatmap for the negative sentence the film is not good at all highlights the fact that the Layer 2 head has increased attention weight for the word 'not' from the word 'good.' The attention heat map for the intensifier sentence it is very rarely funny or exciting shows similar cross-token attention between degree modifiers and the adjectives they modify.

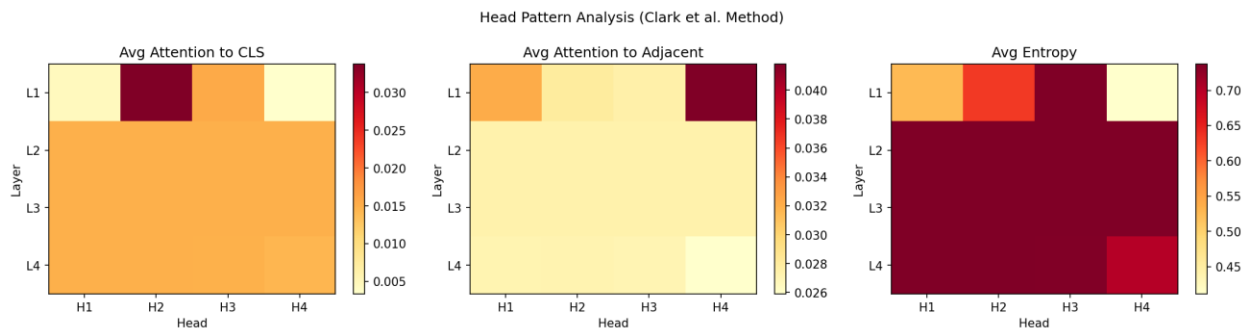


Figure 13. Summary of Head Pattern with Average Attention for CLS token, Adjacent Token Attention, and Entropy across all 16 heads (4 layers $\times$ 4 heads) using Clark et al. (2019) shallow approach on 20 validation batches.

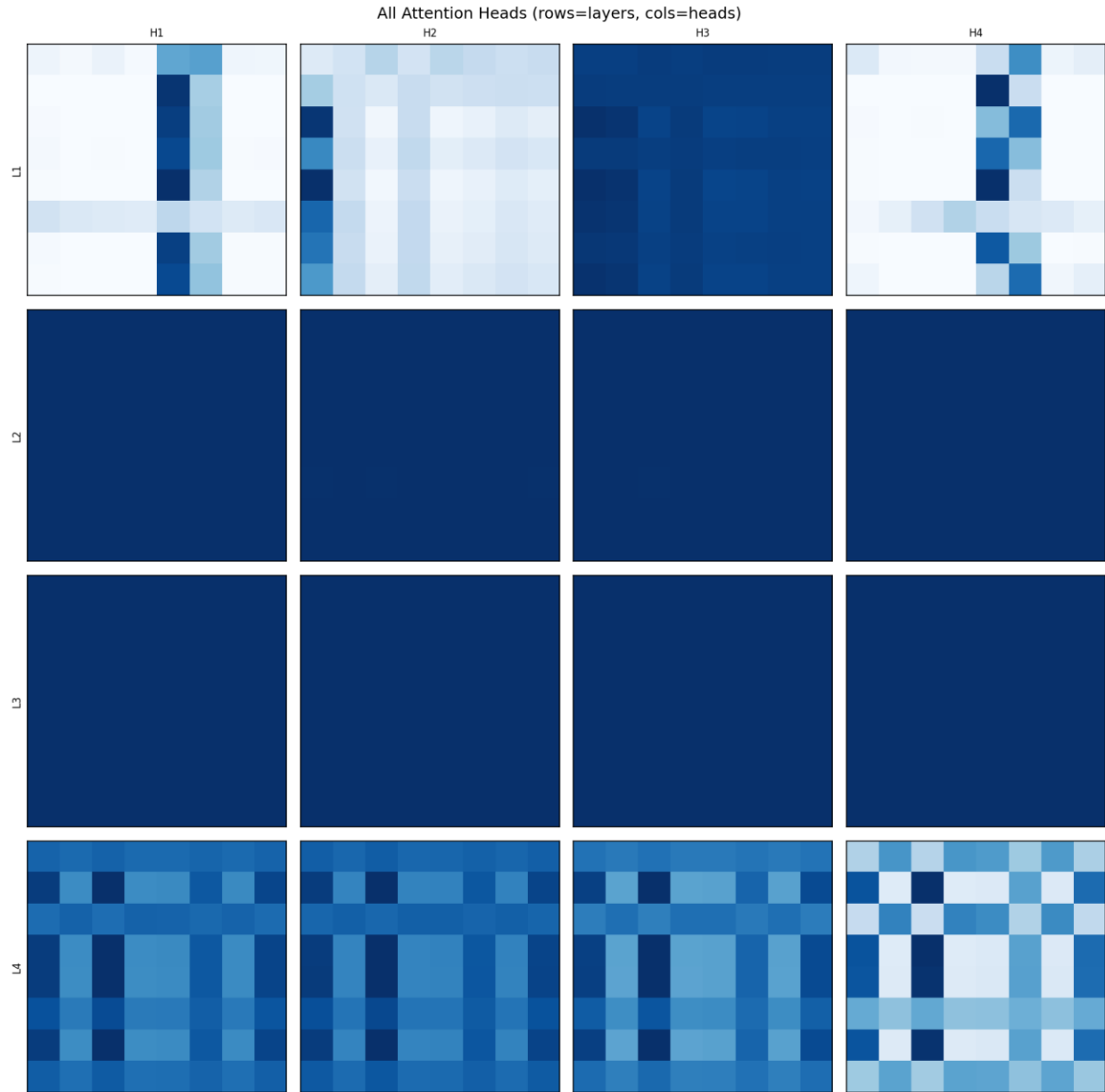


Figure 14. Attention weight heat maps for all 16 heads on the sentence the film is not good at all. Each cell shows the attention weight matrix for one head (rows = query positions, columns = key positions).

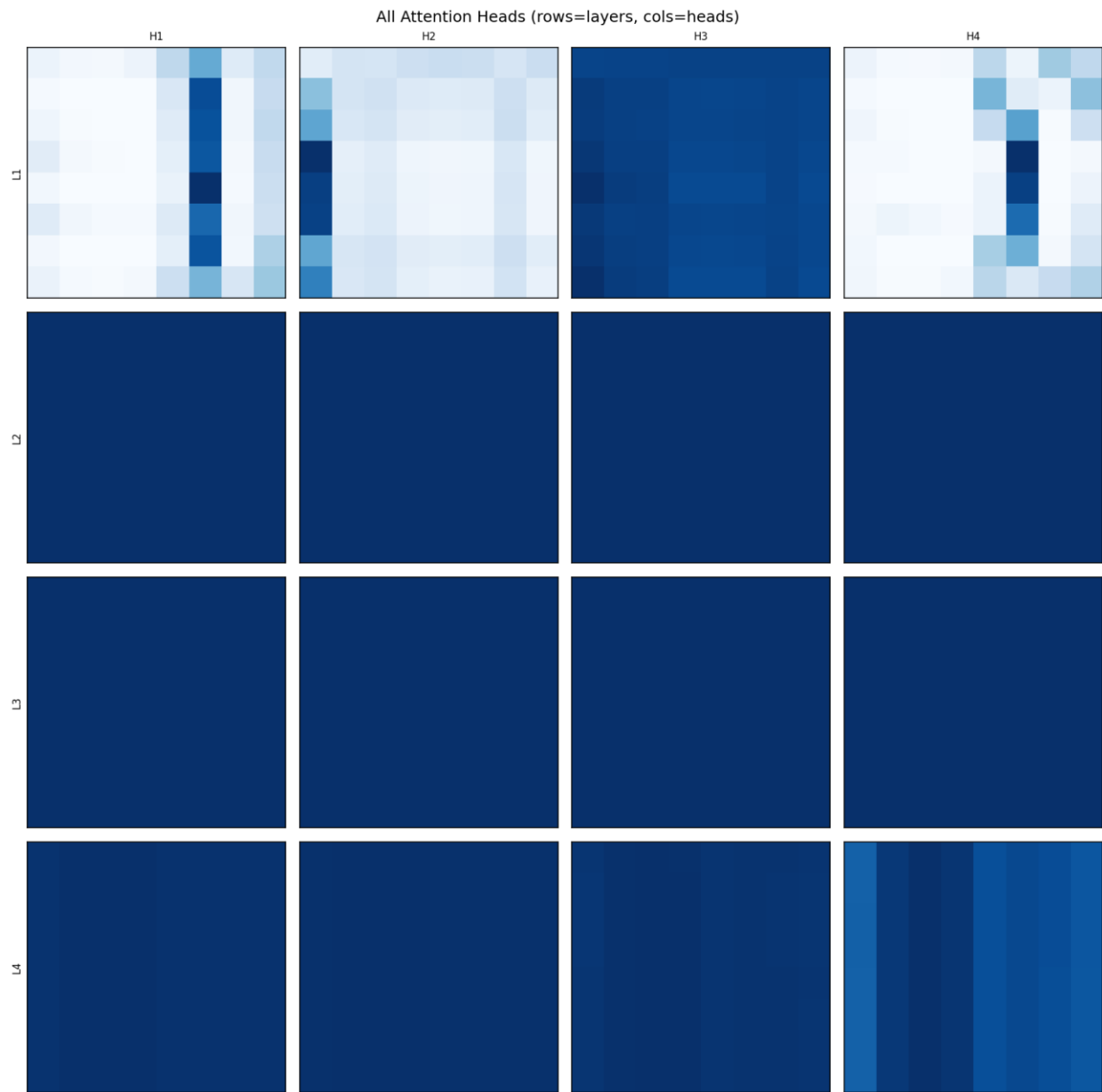


Figure 15. Attention weight heat maps for all 16 heads on the sentence *it is very rarely funny or exciting*.

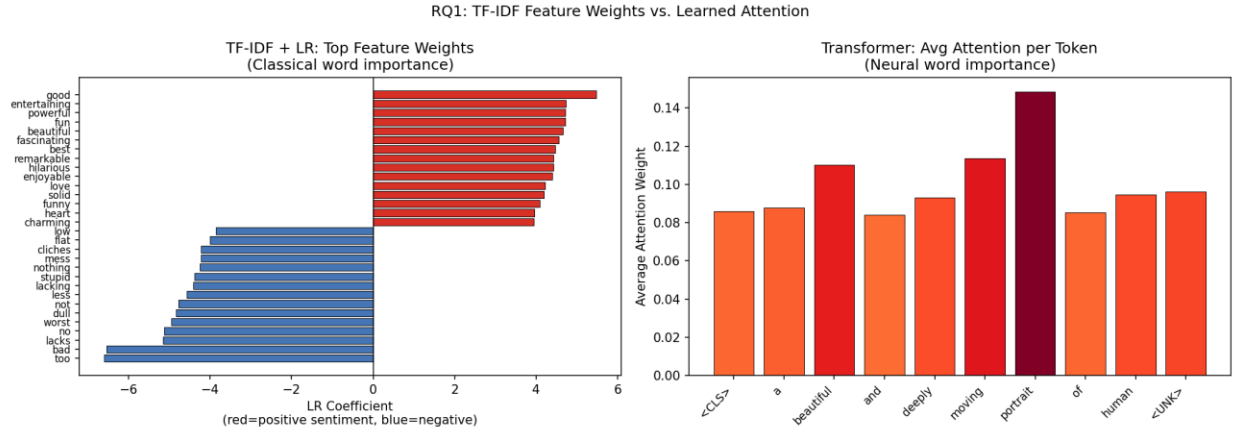


Figure 16. Comparison of TF-IDF logistic regression feature weights (left, static corpus-level coefficients) against Transformer average CLS attention weights (right, context-dependent per-token scores) for a positive-sentiment example sentence.

Tables 4 and 5, and Figures 17 and 18 present the findings of the attention validation experiment. The Spearman rank correlation between average CLS attention weights and gradient saliency scores is 0.2202 at Layer 1, 0.0824 at Layer 2, 0.1457 at Layer 3, and 0.1011 at Layer 4, with an average correlation of 0.2524 across all layers. All correlations have positive values, suggesting that there is partial congruence between attention and gradient saliency in their assessment of which tokens contribute the most to the model’s decisions, albeit not full agreement since the correlation coefficients have only moderate values. It is worth noting that Layer 1, the earliest layer, yields the strongest correlation, whereas Layer 2 produces the weakest. This observation can be attributed to the substantial role played by Layer 2 in compositional processing, as indicated by the decrease in negation probing performance.

Metric	Layer 1	Layer 2	Layer 3	Layer 4
Mean Spearman ρ (n=200)	+0.2202	+0.0824	+0.1457	+0.1011
Overall mean $\rho = 0.2524$				

Table 4: Spearman rank correlation coefficient for CLS attention weight averages and gradient saliency (n=200 validation samples). The correlations in all layers are positive, implying some level of agreement between the two methods. Layer 2 yields the lowest correlation score, which is appropriate since it performs compositionality tasks.

K	Orig. Acc	Attn Removal	Rand Removal	Δ Attn	Ratio (Δ attn / Δ rand)
1	78.50%	72.00% (−6.50%)	78.20% (−0.30%)	−6.50%	21.67×
2	78.50%	65.00% (−13.50%)	76.40% (−2.10%)	−13.50%	6.43×
3	78.50%	68.00% (−10.50%)	76.05% (−2.45%)	−10.50%	4.29×

Table 5. Token deletion faithfulness experiment (sample size = 200). Δ Attn = difference in accuracy after deleting top-K attended tokens. Accuracy for random token deletions is obtained from the average of 10 random runs per sample. Ratio is the relative importance of attention tokens compared to random tokens.

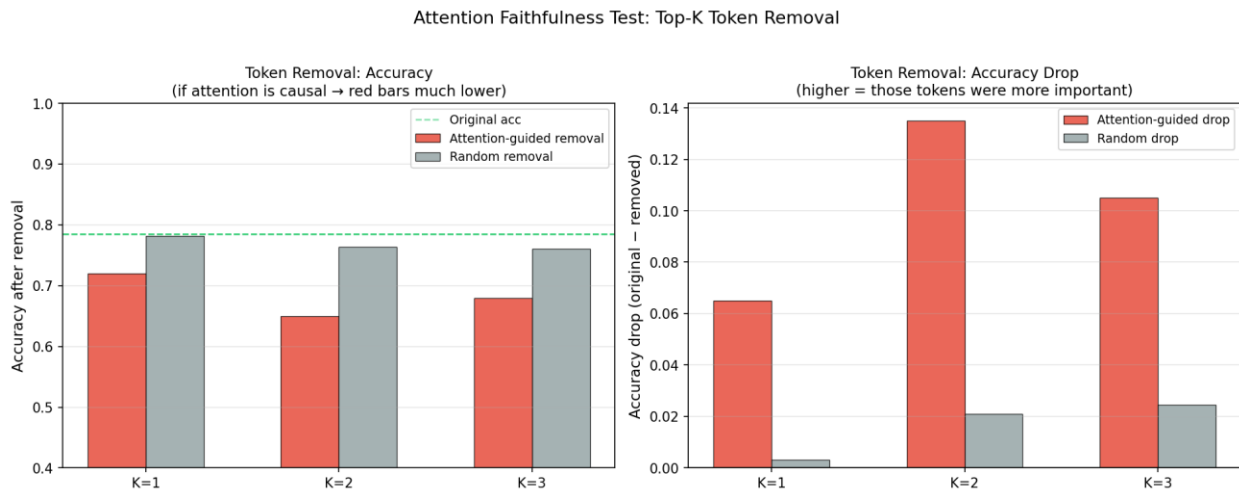


Figure 17. Token removal faithfulness results. Left: absolute accuracy after removal. Right: accuracy

drop. Attention-guided removal (red) causes substantially larger accuracy drops than random removal (gray) at all values of K .

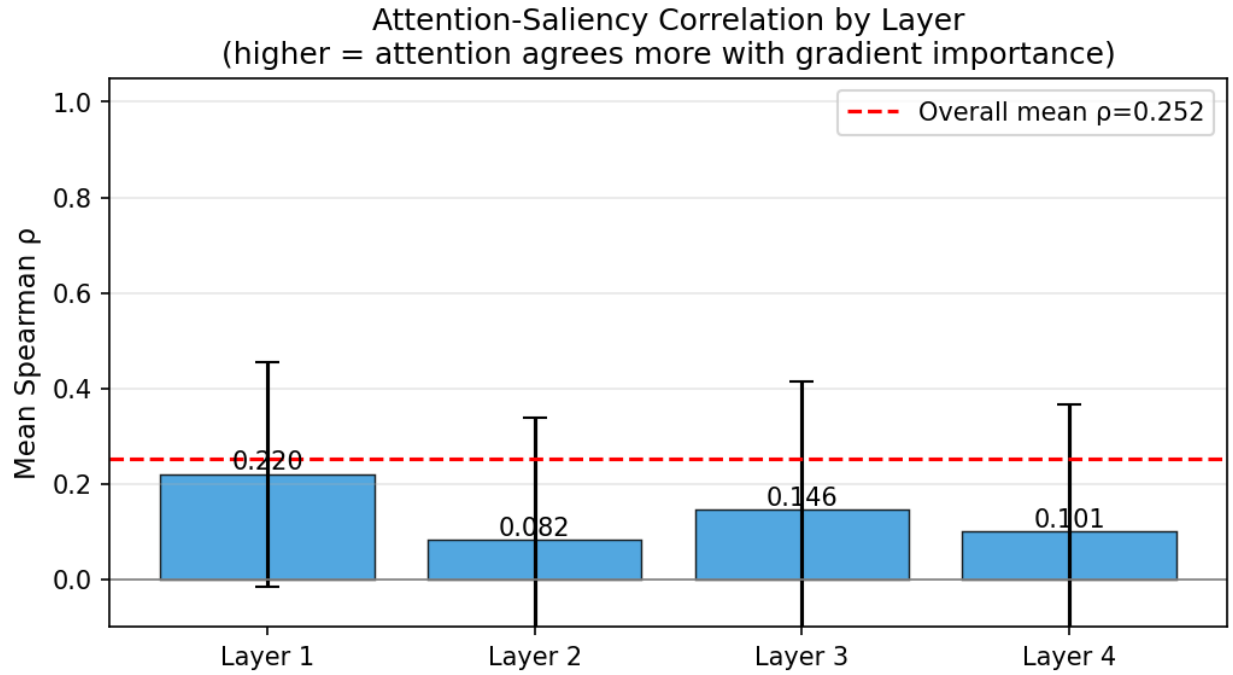


Figure 18. Mean Spearman rank correlation between attention weights and gradient saliency per encoder layer with standard deviation error bars across 200 validation examples.

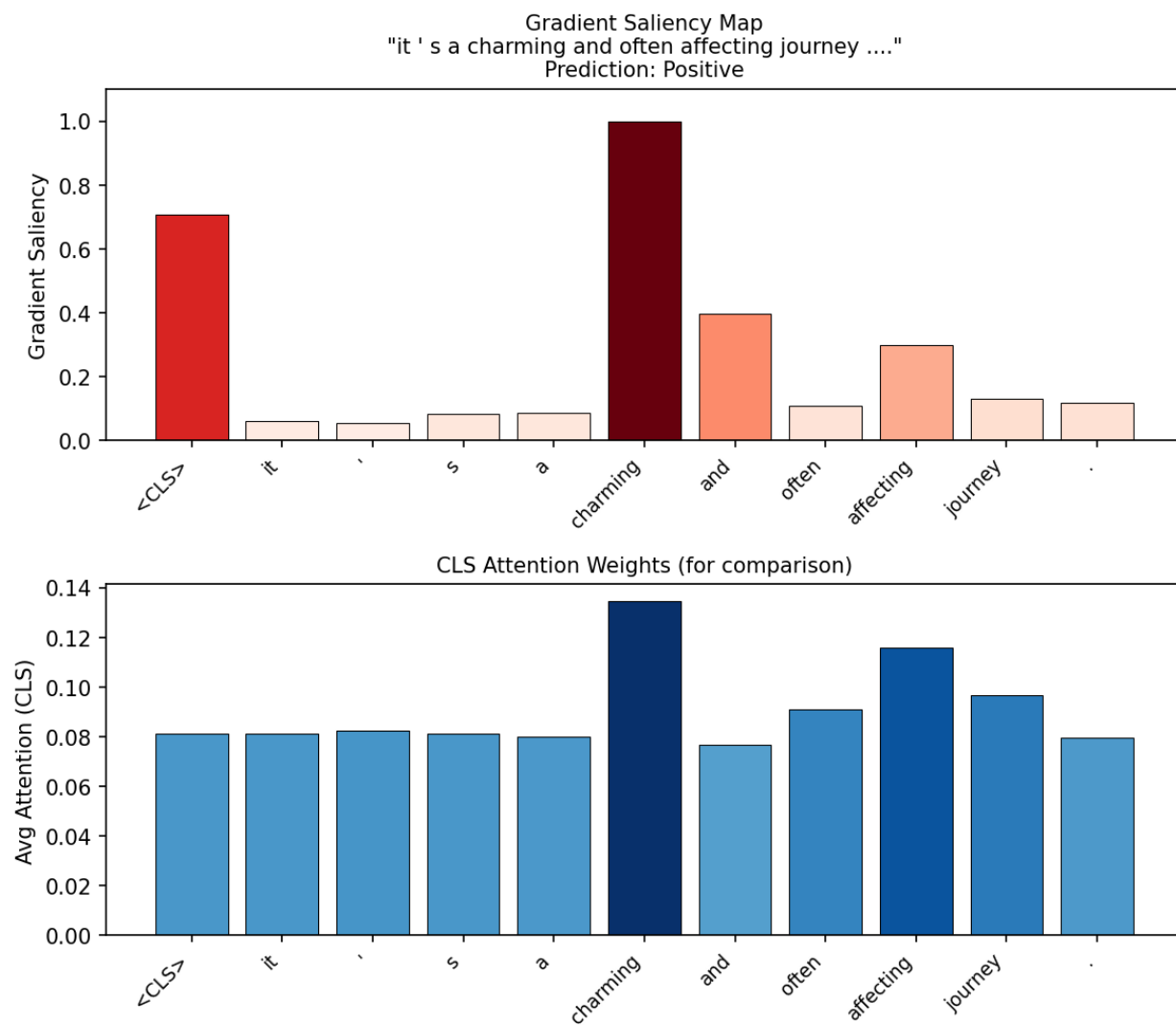


Figure 19. Representative gradient saliency map for a positive validation example. Upper panel shows gradient $\times$ embedding saliency scores per token; lower panel shows average CLS attention weights for comparison.

Conclusions

This paper demonstrates that a Transformer encoder, when designed from scratch with all its components derived from mathematical first principles using only basic PyTorch functions, can perform at par with classic feature-based approaches in the task of sentiment analysis on SST-2 while creating an interpretable internal representation which is not possible for such classic approaches. The BiLSTM model obtains the best accuracy of 82.57%, followed by the TF-IDF approach with an accuracy of 80.96% and the Transformer approach with an accuracy of 80.85%. The second research question pertains to comparing the performance of attention-based representations to that of the TF-IDF feature weights. The Transformer performs 0.11 percentage points worse in terms of accuracy relative to TF-IDF, a statistically meaningless difference. What makes the comparison interesting is the different representations of word importance by these two approaches. Whereas TF-IDF logistic regression applies an equal positive weight to the word good across all samples, making it interpret two different phrases (not good at all and good movie) equally, the Transformer learns context-aware word weighting indicated by the presence of attention links between good and not in Layer 2. The fact that TF-IDF has an inherent positive class bias in terms of recall (i.e., lower negative recall compared to positive recall) also proves that TF-IDF remains anchored in the word frequency. On the other hand, the more balanced per-class metrics achieved by the Transformer demonstrate that attention-based word representations are somewhat unaffected by word frequency statistics. The better AUC-ROC score obtained by TF-IDF logistic regression (0.9065 compared to 0.8869) is associated with better probability calibration in this task.

Secondly, the focus was on attention head specialization. The head pattern analysis reveals the presence of layer-wise specialization in accordance with the results reported by Clark et al. (2019)

in spite of the drastically smaller size and completely untrained nature of this particular model. Head specialization in the first layer is manifested in lower entropy and higher attention to adjacent tokens that act as local context accumulators similar to a windowed convolution. With each subsequent layer, more global and sentence-specific attention patterns emerge, along with increasingly greater amounts of attention being paid to the classification token. This corresponds to the construction of the sentence-level sentiment representation. The discovered patterns hold true for all five sentences examined and also 20 aggregated validation batches.

On the causality question concerning the importance of attention weights, the answers are more complex, based on the threefold examination. In terms of the probing analysis, it becomes clear that sentiment polarity is encoded well since the very beginning in Layer 1 (80.0%), remains constant throughout all layers, and there is also a pronounced monotonous decrease in the accuracy for negation word identification from 62.3% in Layer 1 to 47.1% in Layer 4. The latter can be interpreted as proof that attention actively shifts the negation information from the negation token location to the modified ones. According to the Spearman correlation, attention and gradient saliency have partial agreement (average correlation of 0.2524), where Layer 2 exhibits the weakest correlation because of its task of redistributing the compositionally negated elements leading to relational attention. The token removal test presents some causative evidence in this regard, as the removal of the token with the highest average attention leads to an accuracy decrease of 6.5 percentage points, while the deletion of a randomly selected token leads to a reduction of only 0.3 percentage points. Thus, the difference between these two results is 21.67, when K equals 1. At K equal to 2 and K equal to 3, this difference amounts to 6.43 and 4.29, respectively. Overall, attention tokens have been proven much more causative than randomly selected tokens. This

supports the assertion of Wiegrefe & Pinter (2019) about the possible causal influence of attention.

In terms of the hierarchy of component significance, the ablation experiment yields an unambiguous ranking. Residual connections have no alternative as architectural elements; the absence of them results in total failure of the learning process down to chance level with MCC at 0.000 and a number of true negatives at zero. The practical justification of the hypothesis about the necessity of skip connections as a solution to vanishing gradients is provided by the empirical evidence of such an effect after their elimination. Normalization is a vital element to ensure training stability but does not contribute to final performance in any significant way, leading to dramatic instability during epoch 1 with eventual recovery to 81.31%. Multi-head attention shows superiority to its single-head alternative in terms of both accuracy and faster convergence (11 epochs versus 20 epochs). Positional embedding is not a requirement for SST-2 task, resulting in the marginal gain of 0.46 points of accuracy.

The scaling experiments revealed an unexpected failure case where the medium-size model, having $d_{\text{model}} = 128$ with four layers, resulted in performing by chance, just like the no-residual ablation case, whereas the small-size model, having $d_{\text{model}} = 64$ with two layers, attained an accuracy of 80.05% and the large-size model, having $d_{\text{model}} = 256$ with four layers, scored 80.96%. The implication drawn from the experiment is that there exists a sensitivity for the combination of model depth, embedding dimensions, and the Vaswani warm-up learning rate schedule, as the latter depends on d_{model} . In the case of a peak learning rate corresponding to $d_{\text{model}} = 128$, it could be unstable to train four layers but not two layers of the same embedding dimension (small model).

Several limitations constrain the scope of this study. The first one is that of data size used during the training process. At 67,349 samples, the Transformer model is not supplied with sufficient data for exploiting its representational superiority over sequential inductive bias models. Initialization using pre-trained word embeddings such as GloVe or fastText would help incorporate lexical information into the model, decreasing the data need and closing the accuracy gap with the BiLSTM. The data size used for evaluating probes is relatively small – only around 174 samples for testing the probe after splitting the 872-sample validation set at 80/20. Future research should explore other linguistic features such as parts of speech and constituents. A more comprehensive scaling experiment should be performed, examining different model sizes and controlling for `d_model` independently from the learning rate. It would also be beneficial to further examine the connection between head entropy and functional specialization.

Works Cited

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